6.1 Experiment 1: complete particle–mesh operator action

Experiment 1 asks whether the operator that the integrator actually applies, $X \mapsto \Gamma_h X$, reproduces the periodic friction operator Γ of the screened longitudinal kernel $K(\mathbf{r}) = e^{-\kappa r} r^{-1} P(\mathbf{r})$, $\kappa = 1$, with the error predicted by the choice of (ξ, r_c, k_c, h, p). We follow the hierarchy used for the Landau particle method: static configurations first, so that no time-discretization error enters; then each source of error on its own; then the complete action. The model is not changed for the test. The friction is fully periodic, the $k = 0$ Fourier mode is retained in both the reference and the particle–mesh operator, and there is no free-space truncation and no energy null space. The only null space is the translation space $\mathcal{T} = \mathrm{span}\{T_\alpha\}_{\alpha=1}^{3}$, $T_\alpha = \mathbf{1}_N \otimes \mathbf{e}_\alpha$.

Why small dense-reference systems. Every error in this experiment is measured against a dense $3N \times 3N$ matrix Γ_{ref} and its full spectrum. The reference therefore has to be assembled pair by pair over many periodic images, and it has to be diagonalized. This restricts the experiment to $N \leq 256$. These sizes are chosen to make the reference computable; they are not a regime of interest in their own right. The two original test boxes are case A ($N = 30$, $L = 6$) and case B ($N = 40$, $L = 10$). Here L is the edge length of the periodic cube $[0, L)^3$, not a number of boxes. Their number densities are $n = N/L^3 = 0.139$ and 0.040, and $\kappa L = 6$ and 10. Both are single draws of independent uniform positions (seeds 7 and 13), with no minimum distance imposed; the closest pairs are at 0.395 and 0.589. Case B was taken large enough that the smallest nonzero lattice wave number $2\pi/L$ lies below the wave number at which the longitudinal eigenvalue of $\hat{K}^L$ changes sign. No physical argument selected $L = 6$ for case A: it is a small box in which the dense reference is cheap. The parameter sets used later in the paper were accepted when they met their tolerance on these two boxes. Two boxes are not evidence that the same parameters hold for other configurations, and we use them only as the starting point.

Configurations. Table 1 lists all configurations, including the shortest periodic (minimum-image) pair distance $d_{\min}$ of each. Since $\kappa = 1$, the column L also gives κL. The main set has $N = 128$ at the density of case A, $n_A = 30/216$, so $L = 9.733$. It contains three families, each with three independent seeds:

- *uniform*: independent uniform positions with no minimum distance;
- *perturbed lattice*: a bcc lattice of 4^3 cells, every site displaced by an isotropic Gaussian with $\sigma = 0.1\, d_{\mathrm{nn}}$ ($d_{\mathrm{nn}} = 2.107$ is the nearest-neighbour distance), plus a random global shift;
- *clustered*: $N/16$ cluster centres drawn uniformly, and 16 particles per cluster drawn from an isotropic Gaussian of width $\sigma_c = 0.5$ around their centre.

For the clustered family, a draw is rejected if it comes closer than 0.25 to a particle already placed. This avoids near-coincident pairs, whose $1/r$ kernel would make $\|\Gamma_{\mathrm{ref}}\|_2$ arbitrarily large. The realised $d_{\min}$ is 0.251–0.255, so the constraint is active. For the uniform family, which has no constraint, $d_{\min}$ is 0.142–0.411. One configuration per family at $N = 256$ (with $L = 12.26$ at the same density) is a smaller representative check. All random numbers come from seeded generators; the seeds and generator parameters are stored with the positions.

Two box-length sets are kept apart, because they change different things.

- In the *fixed-N* set, $N = 64$ and $L \in \{4, 6, 8, 10, 12, 16\}$. Changing L therefore also changes the density, from $n = 1$ to 0.0156.

Table 1: Static configurations of Experiment 1. L is the edge length of the periodic cube, $n = N/L^3$ the number density, $d_{\min}$ the smallest periodic pair distance (range over seeds), $\kappa = 1$. A and B are the original two dense-reference boxes.

set	generation	N	$L = \kappa L$	$n = N/L^3$	$d_{\min}$	seeds
A	uniform, seed 7	30	6	0.139	0.395	1
B	uniform, seed 13	40	10	0.040	0.589	1
U128	uniform i.i.d.	128	9.73	0.139	0.142–0.411	3
L128	bcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	128	9.73	0.139	1.208–1.366	3
C128	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	128	9.73	0.139	0.251–0.255	3
U256	uniform i.i.d.	256	12.3	0.139	0.148	1
L256	fcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	256	12.3	0.139	1.011	1
C256	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	256	12.3	0.139	0.259	1
box-N	uniform i.i.d.	64	4–16 (6 values)	0.016–1.00	0.090–0.723	3
box-n	uniform i.i.d.	30–240	6–12 (4 values)	0.139	0.055–0.924	3

- In the *fixed-density* set, $n = n_A$ and $L \in \{6, 8, 10, 12\}$, so $N = 30, 71, 139, 240$.

Each set uses three uniform seeds per L.

Reference and its convergence. Γ_{ref} is built from the unsplit kernel by a direct periodic lattice sum,

$$(\Gamma_{\mathrm{ref}} X)_i = \sum_{j \neq i} \sum_{\mathbf{m} \in \mathbb{Z}^3,\ |\mathbf{r}_{ij} + \mathbf{m}L| \leq R} K(\mathbf{r}_{ij} + \mathbf{m}L)\,(X_i - X_j).$$

It uses neither a minimum-image approximation nor any splitting, and because nothing is removed from the sum the $k = 0$ mode is included automatically. The image radius was increased through $R = 15, 20, \ldots, 45$. Over all 44 configurations, the largest change relative to $R = 45$ is 2.5×10^{-12} at $R = 30$, 2.0×10^{-14} at $R = 35$, and 4.3×10^{-16} at $R = 40$, measured in $\|\cdot\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$. This change follows the tail $e^{-\kappa R}$ until it reaches round-off. We take $R = 45$ as the reference. Its uncertainty, below 10^{-15}, is five orders of magnitude below the smallest error tested below (2.3×10^{-10}). The reference is symmetric to 1.0×10^{-16} and annihilates the translations to 1.9×10^{-16}, both relative.

Split identity and coefficient accuracy. On nonzero separations, θ_S is evaluated in closed form and θ_L from its own s-integral (not as $e^{-\kappa r}/r - \theta_S$). The identity $\theta_S + \theta_L = e^{-\kappa r}/r$ (draft Eqs. (7)–(8)) then holds to 2.5×10^{-15} relative error. This was checked for $\xi \in \{0.5, 0.7, 1, 1.4, 2\}$, on a radial grid $10^{-3} \leq r \leq 20$, and on every pair image with $r \leq 12$ of the main and legacy configurations.

At operator level we combine two independent constructions:

- Γ_S from the real-space lattice sum of θ_S;
- Γ_L from the exact lattice Fourier sum $V^{-1} \sum_{|\mathbf{k}| \leq k_c} \hat{K}^L(\mathbf{k}) \cos(\mathbf{k} \cdot \mathbf{r})$, with $k_c = 2\xi \cdot 6.5$, including $\mathbf{k} = 0$.

Their sum reproduces Γ_{ref} to between 6.7×10^{-15} and 2.2×10^{-13} (Fig. 1a). The largest values occur for the perturbed lattices, whose $\|\Gamma_{\mathrm{ref}}\|_2$ is smallest. This level is a floor of the Fourier lattice sum in double precision, not of the split. Accumulating the same sum in 80-bit extended precision leaves it unchanged, so it comes from the float64 rounding of the inputs (coefficients, positions, wave vectors). The coefficients themselves are accurate: with every Gauss–Legendre rule refined

(twice the panels, 30 nodes per panel), $A(k)$, $B(k)$, θ_L and θ_S change by at most 1.6×10^{-15} relative to $\|\hat{K}^L(k)\|_2$ or to $e^{-\kappa r}/r$, for $\xi \le 2$ and $k \le 14\xi$.

Real-space and Fourier cutoffs. Each truncation is measured with everything else exact:

- $\|\Gamma_S^{r_c} - \Gamma_S\|_2$, where Γ_S is the converged real-space lattice sum;
- $\|\Gamma_L^{k_c} - \Gamma_L\|_2$, where $\Gamma_L^{k_c}$ is the exact lattice Fourier sum over $|\mathbf{k}| \le k_c$.

Both are normalized by $\|\Gamma_{\mathrm{ref}}\|_2$ (Fig. 1b,c). We fit $\log E = a - cx^2$ with $x = \xi r_c$ or $x = k_c/(2\xi)$, using only $x \ge 1$ and $E \ge 10^{-12}$, so that the round-off plateau is excluded.

At $\xi = 0.7$ the fitted Gaussian rates are:

- real space: median $c = 0.99$, range 0.98–1.03 over the eleven main and legacy configurations;
- Fourier space: median $c = 0.97$, range 0.96–1.00.

At $\xi = 1$ the medians are 0.99 and 0.97. The $N = 256$ configurations fall in the same ranges. Both rates are biased by the algebraic prefactor of the tails (draft Eq. (17), $E \propto x\, e^{-x^2}$): dividing by x before the fit gives medians of 1.05 (real space) and 1.03 (Fourier). We do not interpret c as a measured universal exponent.

The plateaus are $\le 8 \times 10^{-16}$ in real space and $\le 1.6 \times 10^{-13}$ in Fourier space. The latter is the lattice-sum floor discussed above. The finite particle set makes the curves staircases at small cutoffs:

- on the perturbed lattices, 13 real-space steps with $\xi r_c \le 1.25$ admit no new pair image;
- for case A, 3 Fourier steps with $k_c/(2\xi) \le 1$ admit no new lattice shell.

These points keep their equal errors and are flagged in the data, not smoothed.

Mesh spacing, B-spline order and aliasing. With $\xi = 0.7$ and the production Fourier cutoff $k_c = 5.74$ ($s = 4.1$), we compare the particle–mesh long-range operator with the exact lattice sum over the same modes. The difference $\Gamma_L^{\mathrm{mesh}} - \Gamma_L^{k_c}$ is then pure mesh error, with no truncation in it. The scan covers $p = 3, \ldots, 8$ and nine FFT-friendly meshes from $\eta = k_c h/\pi \approx 0.99$ down to 0.25. Each configuration is evaluated at eight grid shifts: the unshifted grid and seven random global offsets in $[0, h)^3$ (Fig. 1d–f). The same scan was repeated at a converged cutoff, $s = 5.5$ ($e^{-s^2} \approx 7 \times 10^{-14}$), for $p = 4, 6, 8$.

We fit the RMS over shifts on the four finest meshes with error at least 10^{-12}. This threshold excludes the finest meshes for high p at $s = 5.5$, where the error, down to 8.9×10^{-14} for $p = 8$ at $\eta = 0.25$, approaches the $\sim 10^{-13}$ floor of the exact Fourier lattice sum it is compared with. The fit windows therefore lie between $\eta = 0.23$ and 0.77, typically 0.24–0.41. The fitted slopes in h are, as medians with ranges over the eleven main and legacy configurations (for $p = 5, 7$ the three $N = 256$ checks, 5.12–5.39 and 7.44–7.58, fall inside these ranges):

p	slope at $s = 4.1$, median (range)	slope at $s = 5.5$, median
3	3.09 (2.90–3.22)	–
4	4.07 (3.85–4.36)	4.10
5	5.25 (4.88–5.43)	–
6	6.19 (6.13–6.44)	6.25
7	7.47 (7.03–7.66)	–
8	8.50 (8.42–8.88)	8.87

The error is therefore $O(h^p)$ in this range for every family. For $p \geq 5$ the slopes exceed p, by up to 1.0 for $p = 8$. This is consistent with the leading alias image (draft Eqs. (54), (58)): its factor q^p, with $q = \eta/(2-\eta)$ at the cutoff mode, has local exponent between p and $2p/(2-\eta)$ in h, and reaches p only as $\eta \to 0$. Slopes between adjacent meshes are too noisy to show this directly: the meshes differ by 10–25% in h and the shift sample is eight. Down to $\eta \approx 0.25$ the asymptotic constant of the h^p law has not been reached, and we report no such constant.

The phase sensitivity is moderate. Over all scanned meshes, the largest error over shifts is at most 1.30 times the RMS and at most 1.92 times the smallest; the RMS and the maximum decay at the same rate. On the unshifted grid, the complete off-grid decomposition

$$K_h - K_F = V^{-1} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_{ij}} \left[D_h + H_h(\alpha_i^* + \alpha_j + \alpha_i^* \alpha_j) \right]$$

reproduces the particle–mesh pair response to 3×10^{-15} of $\max |K_F|$. The literal spread–FFT–influence–gather response, computed one source particle at a time, agrees with the mode-space pair response to 1.3×10^{-15}. We checked this for $p = 4, 7$ on two meshes for case A, case B and seed 1 of each $N = 128$ family. In the decomposition:

- the linear cross terms $\alpha_i^* + \alpha_j$ between the principal term and the aliases carry 0.996–1.003 of the mesh error (spectral norm);
- the alias–alias term carries at most 0.9%;
- D_h vanishes to 5×10^{-9} of the error, as it should with ideal deconvolution.

An estimate that kept only the alias–alias term would therefore miss the mesh error almost completely.

The complete production action. We now apply the production code path itself: the real-space pair loop, B-spline spreading of the three components, FFT, multiplication by the influence function $\hat{K}^L/|\hat{W}|^2$ on $|\mathbf{k}| \leq k_c$, inverse FFT, adjoint gather, and the degree correction. We use it in two ways:

- applied to the $3N$ unit vectors, it gives Γ_h column by column, without any hand-assembled matrix;
- applied to probe vectors, it gives the action directly.

Three parameter sets were tested (Table 2): the cheapest sets accepted at $\xi = 0.7$ for the tolerances 10^{-5}, 10^{-7} and 10^{-9} on cases A and B (T5, T7, T9). The literal columns agree with an independent mode-space assembly to 1.1×10^{-15}. They also agree with the sum of the separately computed real-space, Fourier and mesh pieces to 1.4×10^{-15}, so the error budget below decomposes the production operator itself. All three sets keep $\mathbf{k} = 0$ and exclude no influence mode.

Figure 2a shows the relative spectral error $\|\Gamma_h - \Gamma_{\text{ref}}\|_2 / \|\Gamma_{\text{ref}}\|_2$. Cases A and B meet their tolerances, as they were selected to. The uniform and clustered configurations at $N = 128$ and 256 meet them with margin. The perturbed lattices do not:

- T5: 1.16–1.29×10^{-5};
- T7: 1.94–2.11×10^{-7} at $N = 128$ and 1.09×10^{-7} at $N = 256$;
- T9: 1.96–2.55×10^{-9}.

Table 2: Parameter sets of the production action test (accepted at $\xi = 0.7$ for the stated tolerance on cases A and B). M is the next $2, 3, 5$-smooth mesh size with $h \le \eta_N \pi / k_c$; the realised $\eta = k_c h / \pi$ is therefore at most η_N and depends on L.

set	ξ	s	$r_c = s/\xi$	$k_c = 2\xi s$	η_N, p	M (A / B / N=128 / N=256)
T5	0.7	3.494	4.991	4.891	0.8, 6	12 / 20 / 20 / 24
T7	0.7	4.100	5.857	5.740	0.7, 7	16 / 27 / 27 / 36
T9	0.7	4.628	6.611	6.479	0.6, 8	24 / 36 / 36 / 45

The excess is up to a factor 2.6.

Figure 2b shows the reason. At T7, the absolute error $\|\Gamma_h - \Gamma_{\text{ref}}\|_2$ lies in 5.7×10^{-8}–5.5×10^{-7} over all 44 configurations, a factor below ten. Meanwhile $\|\Gamma_{\text{ref}}\|_2$ ranges from 0.72 to 35, set by the closest pairs through $\theta(d_{\min}) \approx e^{-d_{\min}}/d_{\min}$. Perturbed lattices have no pair closer than 1.0 and the smallest norms (0.72–0.94). A tolerance stated relative to $\|\Gamma_{\text{ref}}\|_2$ and calibrated on configurations that contain close pairs therefore does not transfer to configurations without them. The parameters control the absolute error; the relative error inherits the configuration dependence of the norm. At T7 the mesh term dominates the budget ($\le 1.5 \times 10^{-7}$ relative), followed by the real-space and Fourier truncations ($\le 1.0 \times 10^{-7}$ and $\le 7 \times 10^{-8}$, largest for the lattices).

The probes are applied through the production matvec (Fig. 2c):

- sixteen mean-free Gaussian vectors;
- the lowest nonzero periodic wave number $2\pi/L$ along each axis, longitudinal and transverse, cosine and sine, projected onto $\mathcal{T}^{\perp}$;
- the three translations.

At T7 the largest relative action errors $\|(\Gamma_h - \Gamma_{\text{ref}})X\| / \|\Gamma_{\text{ref}} X\|$ are 1.7×10^{-7} (Gaussian), 2.7×10^{-7} (longitudinal) and 4.2×10^{-7} (transverse) on the main and legacy sets. These exceed the relative spectral error because the long-wavelength modes are nearly in the null space of a graph Laplacian: $\|\Gamma_{\text{ref}} X\| / (\|\Gamma_{\text{ref}}\|_2 \|X\|)$ is as small as 6.5×10^{-3} (longitudinal) and 2.9×10^{-3} (transverse). For the translations we report operator-normalized residuals instead of dividing by a zero reference action:

- the production $\Gamma_h T_\alpha$ is exactly zero in floating point, because the degree correction cancels the constant field term by term;
- $\|\Gamma_{\text{ref}} T_\alpha\| / (\|\Gamma_{\text{ref}}\|_2 \|T_\alpha\|) \le 1.9 \times 10^{-16}$.

The literal Γ_h is symmetric to 1.0×10^{-16} (relative Frobenius norm). For Experiment 3 we record the smallest eigenvalue on $\mathcal{T}^{\perp}$. It ranges over 8×10^{-4}–1.92 across the configurations, and is changed by Γ_h by at most 4.2×10^{-5} (T5), 1.6×10^{-6} (T7) and 2.6×10^{-8} (T9) relative. In every case $\|\Gamma_h - \Gamma_{\text{ref}}\|_2 < \lambda_{\min}(\Gamma_{\text{ref}}|_{\mathcal{T}^{\perp}})$, with margin at least $0.997\, \lambda_{\min}$.

Box length. At T7, the two box-length sets give different information (Fig. 2d).

At fixed $N = 64$ the density changes with L. The absolute error falls from 5.3×10^{-7} ($L = 4$, $n = 1$) to 1.0×10^{-7} ($L = 16$, $n = 0.016$). This is mainly because the real-space and Fourier truncation errors grow with the number of neighbours inside the kernel range: the absolute Γ_S term grows from 2.8×10^{-8} to 3.2×10^{-7} as n rises from 0.016 to 1, sublinearly in n.

At fixed density n_A the absolute error stays between 1.1 and 2.7×10^{-7} for $L = 6$–12, with no trend in L. Its variation follows the realised η. Rounding M to a $2, 3, 5$-smooth size gives $\eta = 0.61$

at $L = 8$ but 0.68–0.69 at $L = 6, 10, 12$, and the mesh term scales roughly as η^p. The same rounding causes the dip at $L = 8$ in the fixed-N set.

A fixed-N comparison is therefore a density test, and only the fixed-density comparison isolates L; at fixed density no box-length effect beyond the mesh rounding was found. The relative error additionally inherits the configuration-dependent $\|\Gamma_{\mathrm{ref}}\|_2$, as above.

What the experiment supports. The original cases A and B show that the accepted parameter sets meet their relative tolerances on two uniform boxes. On their own they do not support a claim about other configurations. The supplementary experiment adds the following:

1. an independent reference built from the unsplit kernel, with $k = 0$, whose convergence is demonstrated to below 10^{-15};

2. the split identity and the coefficient quadrature at round-off over a range of ξ;

3. the Gaussian decay of both truncations and the $O(h^p)$ mesh law, with consistent rates across three configuration families, two particle numbers and grid shifts, and with the linear alias cross terms identified as the mesh error;

4. a verification of the production spread–FFT–influence–gather action itself, including null-space and symmetry properties.

It also shows a limitation of the original acceptance: tolerances defined relative to $\|\Gamma_{\mathrm{ref}}\|_2$ are configuration dependent and are exceeded by up to a factor 2.6 on perturbed lattices, while the absolute error remains within a factor ten of a configuration-independent level set by the parameters. Parameter selection for production runs should therefore be calibrated on the configuration class with the smallest $\|\Gamma\|_2$, or on an absolute scale.

All results are for $\kappa = 1$, $N \leq 256$ and static configurations. The mesh law is established for $0.25 \lesssim \eta \leq 0.99$, and its asymptotic constant was not reached.

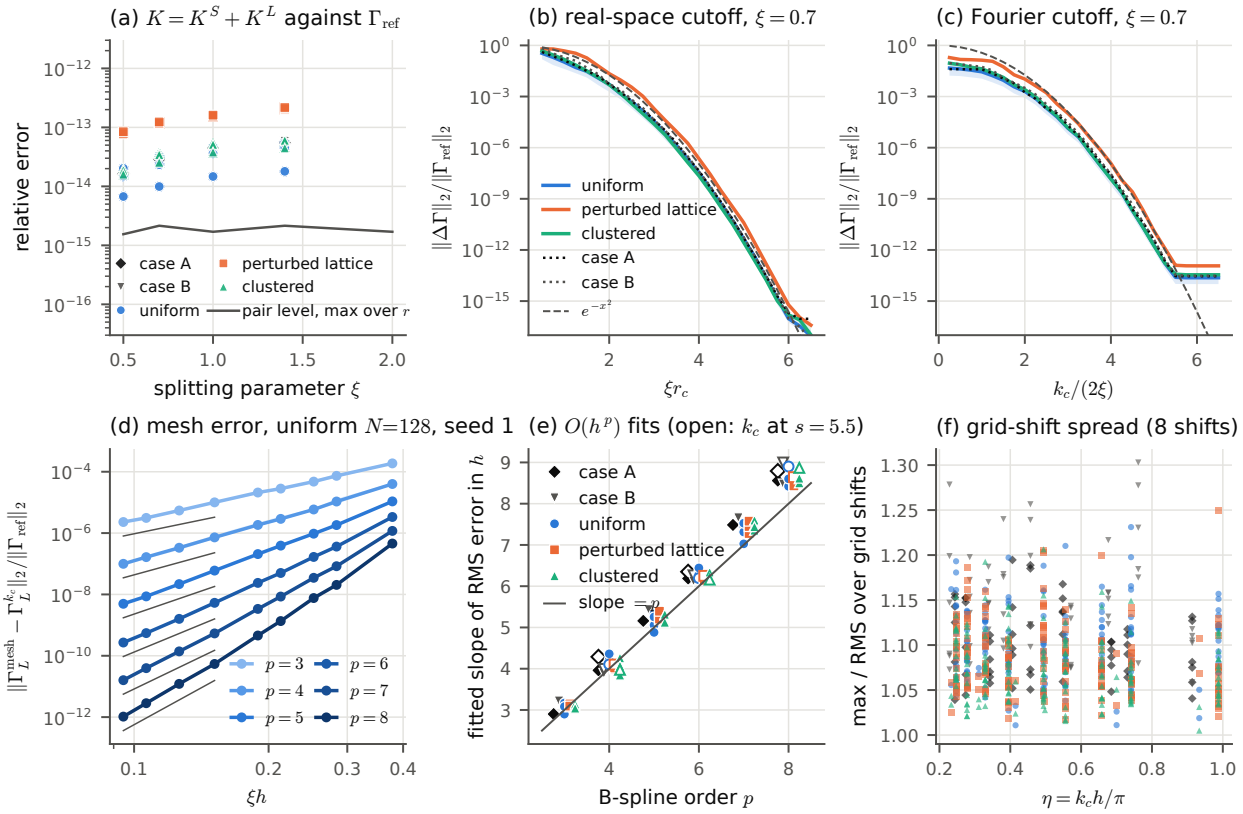


Figure 1: Error sources of Experiment 1, $\kappa = 1$. (a) Split identity: Γ_S (real-space lattice sum) plus Γ_L (exact lattice Fourier sum, $k_c = 2\xi \cdot 6.5$) against Γ_{ref}. The line is the pair-level error $\max_r |\theta_S + \theta_L - e^{-\kappa r}/r|/(e^{-\kappa r}/r)$. (b) Real-space and (c) Fourier truncation at $\xi = 0.7$, with every other part exact. The curves are the median over three seeds and the band is their range; dotted curves are cases A and B; the dashed curve is e^{-x^2}. (d) Mesh error against the exact lattice sum over the same modes at the production $k_c = 5.74$, for uniform $N = 128$, seed 1. The curves are the RMS over eight grid shifts, the bands the range over shifts, and the thin segments have slope p. (e) Fitted slopes in h on the four finest meshes. Filled markers use the production k_c; open markers use k_c at $s = 5.5$. (f) Largest over RMS error across the eight grid shifts, for every mesh, p and configuration.

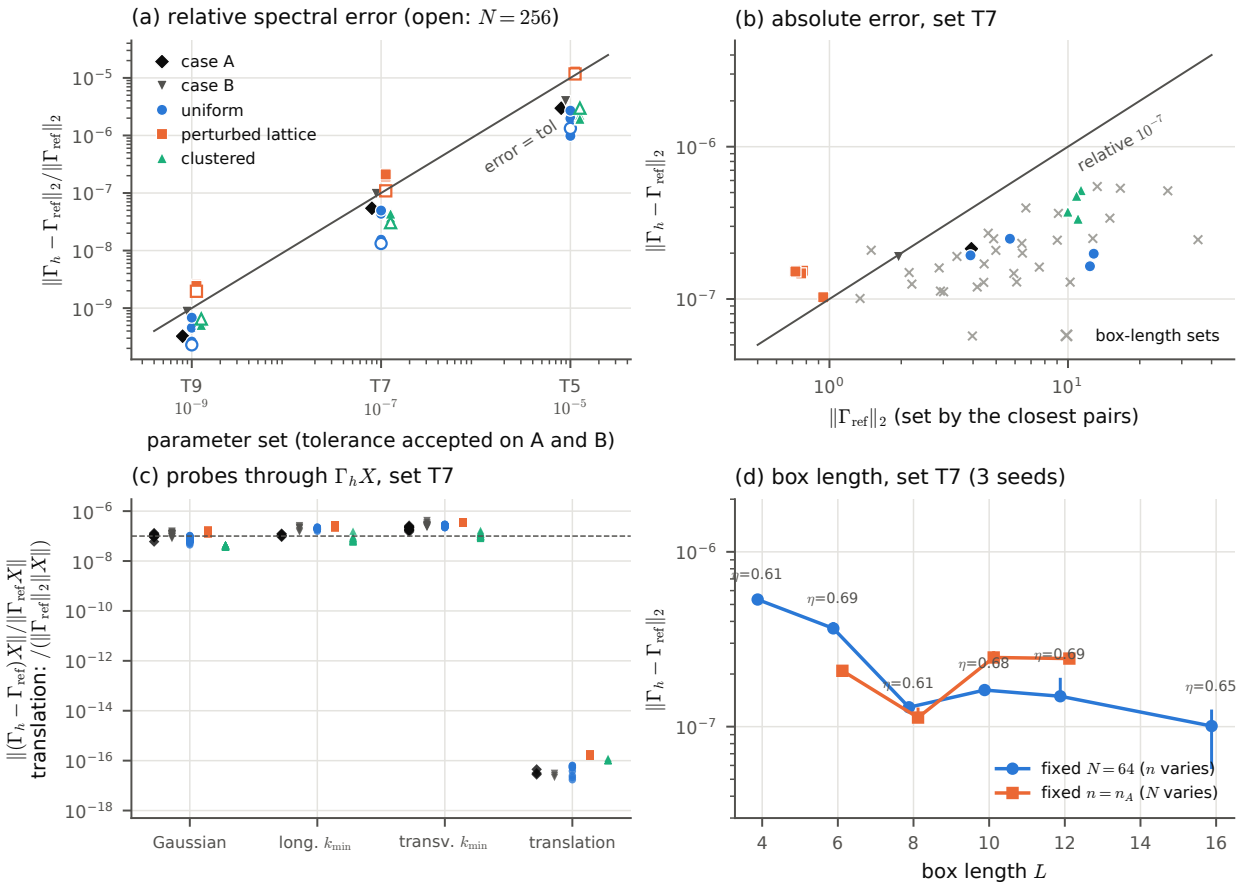


Figure 2: The production action $X \mapsto \Gamma_h X$ against the dense full-periodic reference. (a) Relative spectral error for the parameter sets accepted at 10^{-9}, 10^{-7} and 10^{-5} on cases A and B. Filled markers are $N \leq 128$ and open markers $N = 256$. (b) Absolute error against $\|\Gamma_{\mathrm{ref}}\|_2$ at T7; the diagonal is relative 10^{-7}; crosses are the box-length sets. (c) Probe errors through the production matvec at T7. Translations are given as operator-normalized residuals; the dashed line is 10^{-7}. (d) Absolute error against box length at T7, at fixed $N = 64$ (density varies) and at fixed density n_A (N varies). The realised η is annotated.